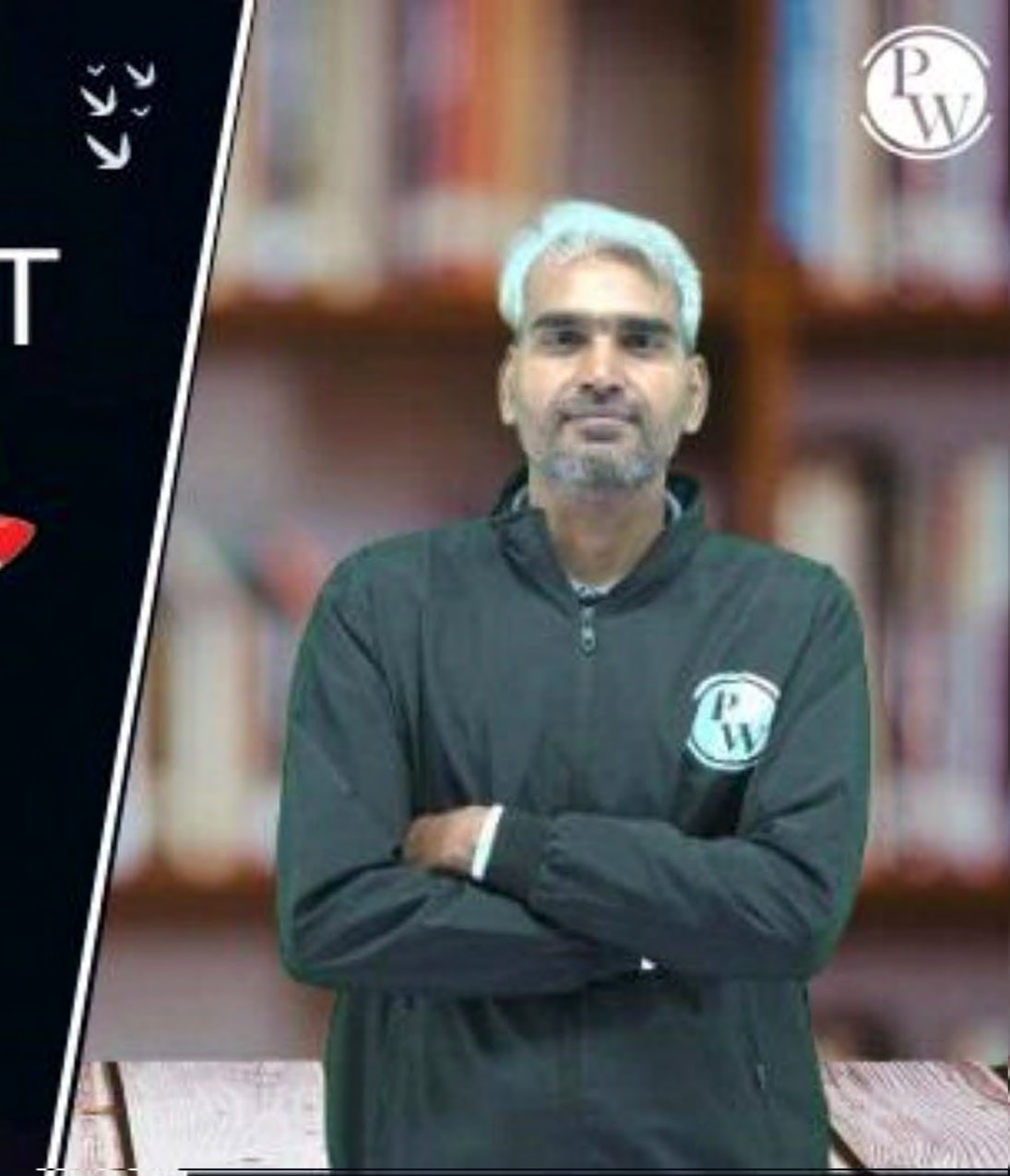
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 15



By- Vishal Sir

Recap of Previous Lecture



- Topic: First normal form (1NF)
- Topic: Redundancy in relation because of FD
- Topic: Second normal form (2NF)
- Topic: Third normal form (3NF)
- Topic: Boyce codd normal form (BCNF)

Slide

Topics to be Covered



Topic — Identifying the normal form of relation

Topic — Decomposition of relation up to BCNF

Slide



Topic : Normalization

- Normalization is the process of decomposing the relation into sub-relations, such that redundancy is reduced or eliminated.
- To specify the different types of redundancy we use the concept of "Normal Forms"



Topic : Normal forms

Boyce Codd Normal Form → introduced the Concept of relational model.

There are various types of normal forms,

1 NF | 2 NF | 3NF | BCNF | 4NF

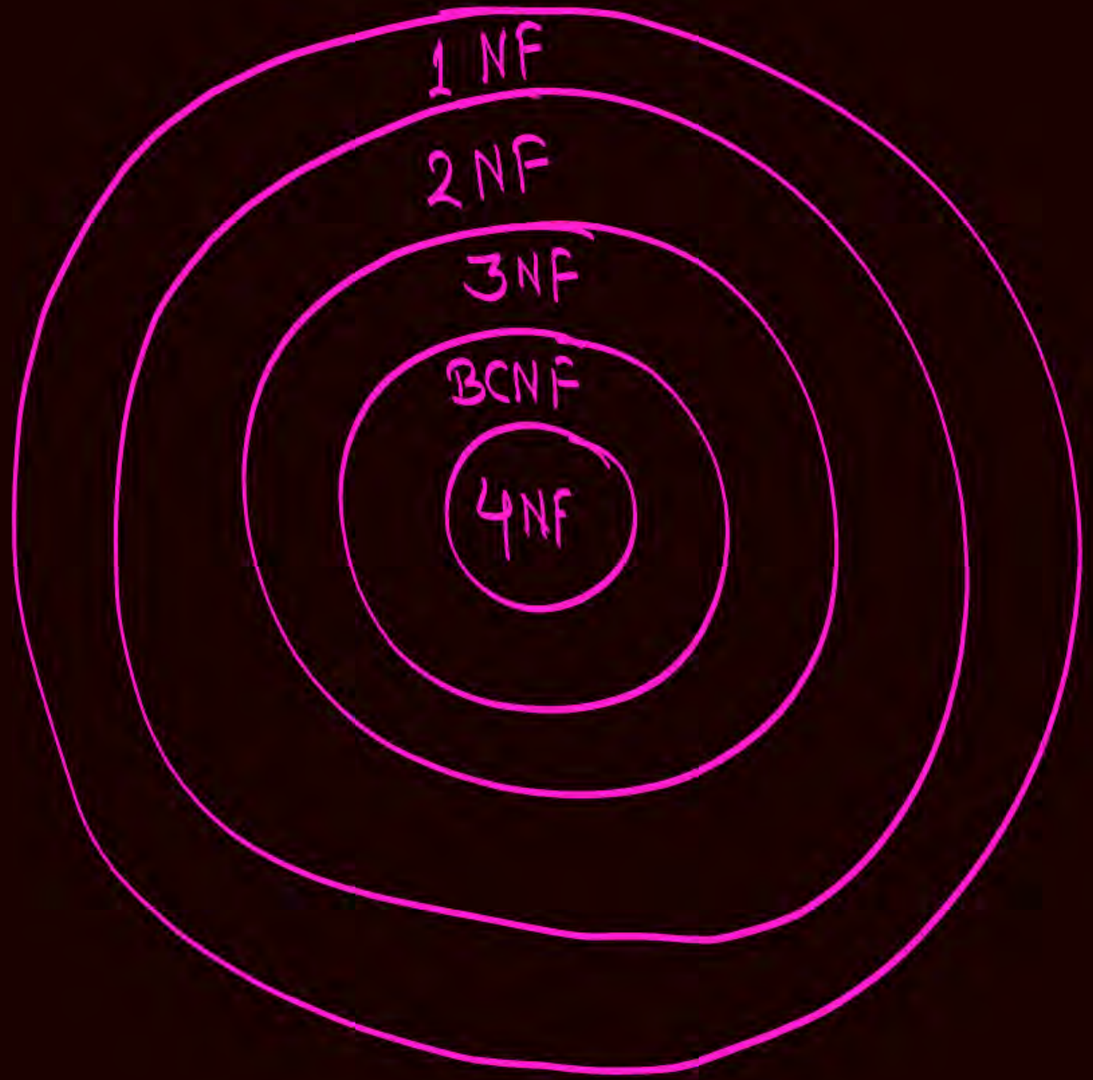
* Upto BCNF we will try to eliminate the redundancies present in the relation because of functional dependencies.
* If a relation is in BCNF, then there will be no redundancy in that relation because of functional dependencies, but a relation which is in BCNF may still suffer from redundancy present in it because of Multi-valued dependencies (MVD)

* 4NF is related to multi-valued dependencies
* In 4NF we will try to eliminate the redundancies present in the relation because of multi-valued dependency.

* Every relation which is in 2NF, is also in 1 NF.
* Every relation which is in 3NF, is also in 2NF and hence also in 1 NF

⋮

and so on



Topic : First normal form (1NF)

A database is in 1NF only if it does not contain any multivalued attribute {i.e. all attributes must be simple and atomic}

Eg:

Sid	Mob_No.
S_1	{12345, 98270}
S_2	{94250, 98930}
S_3	98930

Multi-valued attribute

it is not a relation

Multi-valued attribute is present so Not in 1NF

Convert multi-valued attributes into single valued attributes

Sid	Mob_No.
S_1	12345
S_1	98270
S_2	94250
S_2	98930
S_3	98930

Now, Mob_No. is single valued attribute

No multi-valued attribute exists so it is atleast in 1 NF.

* By default normal form of relation is 1NF.

{i.e, Every relation is at least in 1NF}



Topic : Redundancy in relation because of FD

Rule 1 :- In a functional dependency "$X \rightarrow Y$", if "X" is a Super key, then it does not cause any redundancy in the relation

Rule 2 :- In a functional dependency "$X \rightarrow Y$" if X is not a super key, then it may cause redundancy in the relation

$(A)^+ = \{A, B, C\}$

$\therefore$ A is a S.K

$A \rightarrow BC$

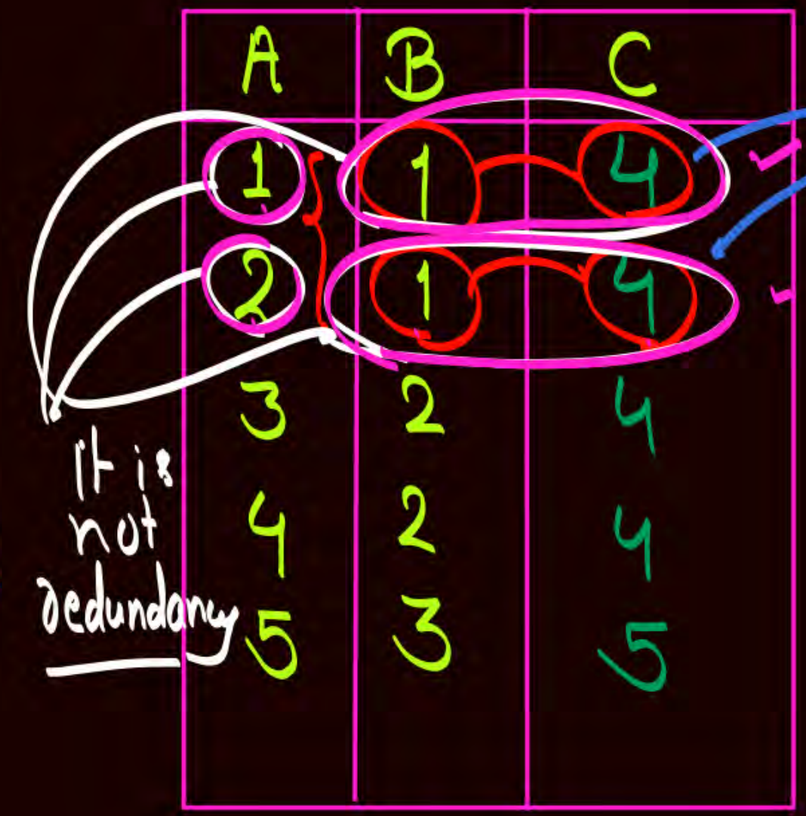
AB is a S.K

$AB \rightarrow C$

$B \rightarrow C$

$(B)^+ = \{B, C\}$

$\therefore$ Not a S.K.



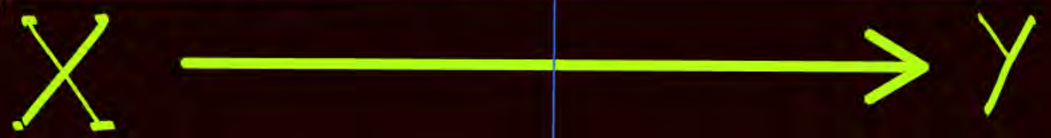
Redundancy ✓

Possible types of non-trivial FDs which may cause redundancy in the relation

Note:- In $X \rightarrow Y$, if X is not a Superkey, then Y can never be a Super key.

∴ If $X \rightarrow Y$ causes redundancy in the relation, then neither L.H.S. nor R.H.S. of that FD can be a Super key.

Possible types of non-trivial FDs which may cause redundancy in the relation



Type ① Proper subset of a C.K → Non-prime attributes ✓

Type ② (Proper Subset of a Candidate Key + Non-prime attributes) → Non-prime attribute

Type ③

Non-prime attributes → Non-prime attributes

Non-Prime Attributes → P.S.C.K. {Such FDs are not Possible}

Proper subset of a C.K → Proper subset of same C.K Such FD is not Possible

Type ④ Proper subset of one C.K → Proper Subset of some other C.K

Type ⑤ (Proper Subset of one Candidate Key + Non-prime attributes) → Proper Subset of some other C.K

Note :-

Normal Form \ FD type	Type 1	Type 2	Type 3	Type 4	Type 5
1 NF	Allowed	Allowed	Allowed	Allowed	Allowed
2 NF	Not allowed	Allowed	Allowed	Allowed	Allowed
3 NF	Not allowed	Not allowed	Not allowed	Allowed	Allowed
BCNF	Not allowed	Not allowed	Not allowed	Not allowed	Not allowed



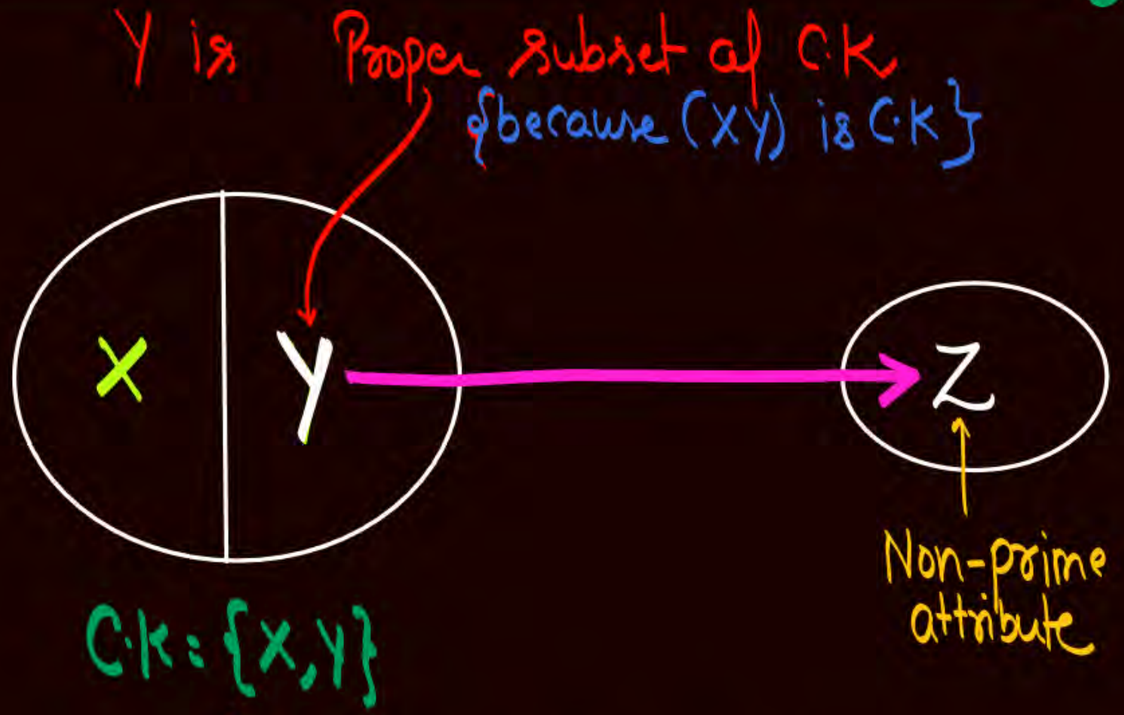
Topic : Second normal form (2NF)

* A relation R is in 2NF if and only if,

① R is in 1NF

and ② Relation R must not contain any Partial dependency

Partial Functional dependency



if $Y \rightarrow Z$ exists, then $XY \rightarrow Z$ is Partial functional dependency

★ Let (XY) is the candidate key and 'Z' is some non-Prime attribute in relation R

① If "Z" is fully dependent on candidate key {i.e, No proper subset of Candidate key can determine "Z"}, then "$XY \rightarrow Z$" is called full functional dependency.

② If Proper subset of Candidate key {X, Y} Can determine "Z", then $XY \rightarrow Z$ is Called Partial Functional dependency

* If type 1 functional dependency { Proper subset of C.k ⟶ Non-prime attribute } exists in the relation, then Partial Functional dependency exist in the relation.

∴ for 2NF,

① Relation must be in 1NF

and ② Relation must not contain any functional dependency of "type 1"



Topic : Third normal form (3NF)

A relation R is in 3NF if and only if,

① R is in 1NF {i.e, No multi-valued attribute}

and ② Every non-trivial functional dependency $X \rightarrow Y$ must have,

X as a Super key

(or)

Y as a prime attribute

→ i.e, Type-1, Type-2, and Type-3 FDs are not allowed.

In $X \rightarrow Y$, if X is a Super key, then there will be no redundancy because of $X \rightarrow Y$, i.e $X \rightarrow Y$ will satisfy BCNF Condn and hence 3NF as well.

Note:- A relation R is in 3NF, only if transitive functional dependency does not exist in relation.

→ A functional dependency is called transitive functional dependency, if non-prime attribute is transitively (not directly) dependent on key.

C.K/S.K. ⟶ "Non-prime Attribute 1"

In this case "Non-prime attribute 1" is directly dependent on key. ∴ it is not a transitive FD ∴ Allowed in 3NF.

Neither a C.K. nor a S.K ⟶ Non-prime Attribute 2

Non-prime attribute is not directly dependent on key. ∴ Transitive FD ∴ Hence Not allowed in 3NF



Topic : Boyce codd normal form (BCNF)

* A relation R is in BCNF if and only if,
  Every non-trivial functional dependency "$X \longrightarrow Y$"
  must have "X" as a Super key.

{ None of the five types of FDs are allowed }

#e.g. Given R(ABCDEF) and

F={AB→CD, D→A, C→E, B→F }

Find the normal form of the relation.

C.K = AB, DB

Prime Attributes : { A, B, D }

Non-Prime Attributes = { C, E, F }

AB → CD

S.K ⇒ ∴ BCNF

D → A

P.S. of one C.K → P.S of another C.K. ∴ 3NF

Type 4 FD

allowed up to 3NF, but not allowed in BCNF

C → E

N.P.A. → N.P.A ∴ 2NF

"Type 3" FD, it is allowed upto 2 NF, but not allowed in 3NF

B → F

P.S.C.K → N.P.A ∴ 1NF

it is "Type 1" FD

allowed in 1NF, but not allowed in 2NF

∴ Least among all the normal form is 1NF.

∴ Highest normal form satisfied by the relation 1NF

Normal form of a relation will be the least of highest normal form satisfied by any of its FD.

#e.g. Given R(ABCDE) and F={AB→C, C→D, B→E }

Find the normal form of the relation, and if relation is not already in BCNF, then decompose the relation up to BCNF.

C.K = (AB)

P.A = {A, B}

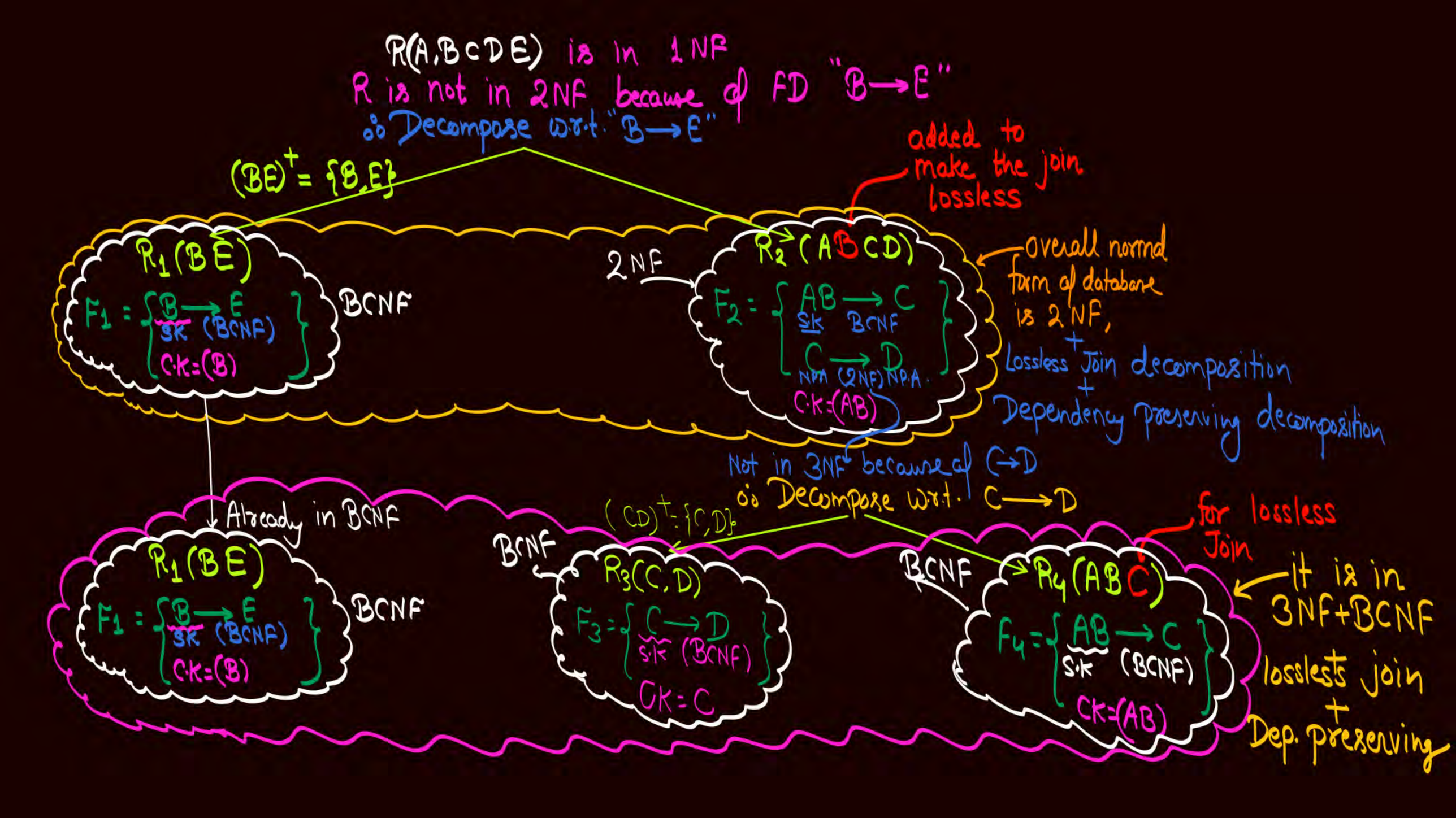
N.P.A = {C, D, E}

F = { AB → C (S.K) (BCNF)

C → D (NP.A → NP.A) (2NF)

B → E (PSCK → N.P.A) (1NF) }

Normal form of Relation is 1NF


R(A,B,C,D,E) is in 1NF
R is not in 2NF because of FD "B → E"
∴ Decompose w.r.t. "B → E"
(BE)+ = {B, E}
added to make the join lossless
R1(B E)
F1 = { B → E (BCNF) SK ; C.K = (B) }
BCNF
2NF
R2(A B C D)
F2 = { AB → C SK BCNF ; C → D NPA (2NF) ; C.K = (AB) }
overall normal form of database is 2NF,
+
Lossless Join decomposition
+
Dependency preserving decomposition
Not in 3NF because of C → D
∴ Decompose w.r.t. C → D
(CD)+ = {C, D}
Already in BCNF
R1(B E)
F1 = { B → E (BCNF) SK ; C.K = (B) }
BCNF
BCNF
R3(C, D)
F3 = { C → D (BCNF) SK ; CK = C }
BCNF
R4(A B C)
F4 = { AB → C (BCNF) S.K ; CK = (AB) }
for lossless Join
it is in 3NF+BCNF
lossless join
+
Dep. preserving

#e.g. Given R(ABCDEF) and F={A→BCDEF, BC→ADEF, D→E , E→F}

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

C.K = {A}, {BC}

P.A = {A, B, C}

N.P.A = {D, E, F}

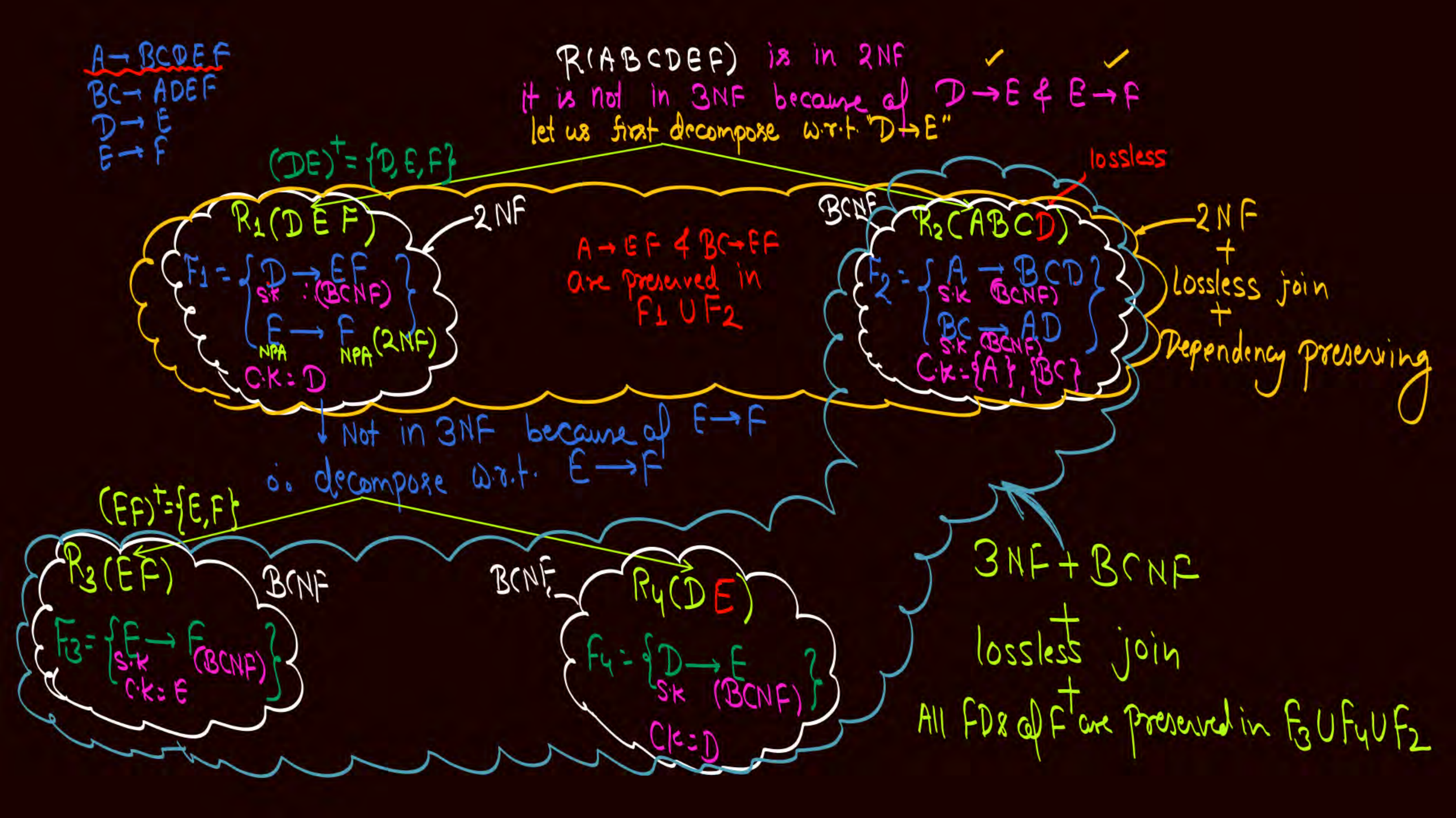
A → BCDEF
S.K ∴ (BCNF)

BC → ADEF
S.K ∴ (BCNF)

D → E
NPA (2NF) NPA.

E → F
NPA (2NF) NPA

Normal Form of Relation is "2NF"


A → BCDEF
BC → ADEF
D → E
E → F
R(ABCDEF) is in 2NF
it is not in 3NF because of D → E & E → F ✓ ✓
let us first decompose w.r.t "D → E"
(DE)+ = {D, E, F}
R1(DEF)
F1 = { D → EF (S.K : BCNF), E → F (NPA NPA (2NF)) }
C.K : D
2NF
A → EF & BC → FF are preserved in F1 ∪ F2
BCNF
R2(ABCD)
F2 = { A → BCD (S.K (BCNF)), BC → AD (S.K (BCNF)) }
C.K :- {A}, {BC}
lossless
2NF + lossless join + Dependency preserving
Not in 3NF because of E → F
i. decompose w.r.t. E → F
(EF)+ = {E, F}
R3(EF)
F3 = { E → F (S.K (BCNF)) }
C.K : E
BCNF
BCNF
R4(DE)
F4 = { D → E (S.K (BCNF)) }
C.K : D
3NF + BCNF + lossless join + All FDs of F+ are preserved in F3 ∪ F4 ∪ F2

H.W: Try the decomposition upto BCNF, while the first decomposition is w.r.t "E → F"

R(ABCDEF) is in 2NF

it is not in 3NF because of D → E & E → F

let us first decompose w.r.t "E → F"

$(EF)^+ = \{EF\}$

$R_1(EF)$

H.W.

#e.g. Given R(ABCD) and F={AB→C, BC→D}

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

H.W.

#e.g. Given R(ABCDEFGHIJ) and F={AB→C, A→DE, B→F, F→GH, D→IJ}

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

#e.g. Given R(ABDLPT) and F={B→PT, T→L, A→D}

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.



2 mins Summary

Topic Identifying the normal form of relation

Topic Decomposition of relation up to BCNF

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THANK - YOU

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